

Retail Customer Intelligence & Sales Analytics System

Business Problem Statement

Business Background :

Retail businesses generate a huge amount of customer shopping data every day. However, simply collecting data is not enough. Companies need to understand how customers shop, what products they prefer, and what factors influence their purchasing decisions.

This project focuses on analyzing customer shopping behavior to identify useful patterns and trends. By studying customer demographics, purchase history, product categories, discounts, subscription status, and seasonal shopping habits, businesses can make better decisions that improve customer satisfaction, increase sales, and build long-term customer loyalty.

Business Objective :

The main goal of this project is to transform raw customer shopping data into meaningful business insights. This project aims to answer the following question:

"How can customer shopping data be used to understand buying behavior, improve customer engagement, and support better business decisions?"

Project Deliverables :

1. Data Preparation & Analysis (Python)

- Import and clean the dataset.
- Handle missing values and prepare the data for analysis.
- Perform Exploratory Data Analysis (EDA).
- Create additional features to improve analysis.

2. Data Analysis (SQL)

- Store the cleaned data in PostgreSQL/ MySQL.
- Write SQL queries to analyze customer behavior.
- Identify customer segments, sales trends, and product performance.

3. Dashboard & Business Insights (Power BI)

- Build an interactive dashboard.
- Visualize important KPIs and customer trends.
- Present insights that support data-driven business decisions.

Expected Outcome

The project helps businesses better understand customer shopping behavior and identify opportunities to improve marketing strategies, customer engagement, and overall sales performance. The insights generated can support smarter and more informed business decisions.